BMB547

Molekylær Data Science

Introduktion

Jonathan Brewer

Hvad er jeres drøm/forestilling om hvad I skal lave under jeres uddannelse og efter ?

Hvorfor er data science vigtigt for jeres uddannelse i life science

•**Analyse af Big Data**: Data science gør det muligt at håndtere store datasæt som genomsekvenser, kliniske forsøg, og epidemiologi.

•**Præcisionsmedicin**: Tilpasser behandling ved at analysere genetik, sygehistorie og livsstil.

•**Lægemiddelopdagelse**: Maskinlæring kan forudsige molekylers opførsel og effektivitet i behandling.

•**Genomanalyse**: Identificerer sygdomsmarkører og genetisk variation via genomdata.

•**Epidemiologi**: Modellerer sygdomsspredning med data science.

•**Billedanalyse**: Analyserer medicinske billeder hurtigere og mere præcist.

•**Kliniske Forsøg**: Hjælper med design og dataanalyse af forsøg.

•**Komplekse Systemer**: Forstår systemer fra molekyler til økosystemer via data science.

Formålet ved kurset

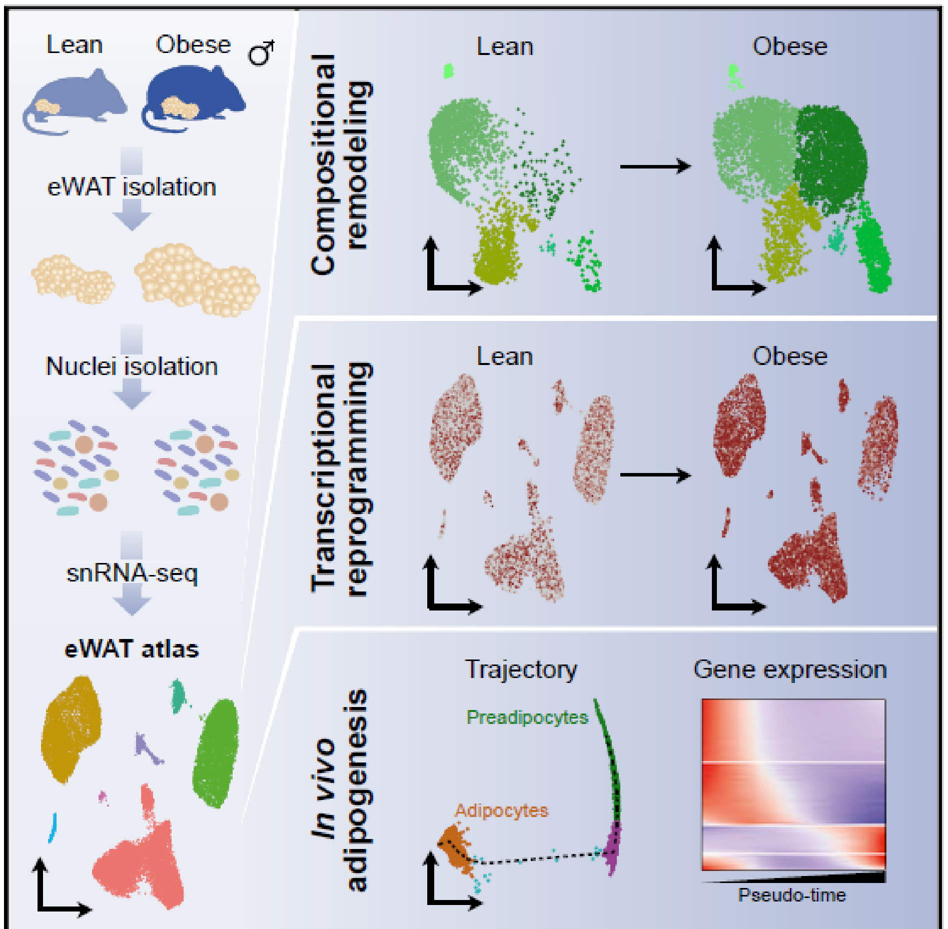
- Lære jer nødvendig matematik og introducere data science kompetencer som i vil have bruge for i løbet af jeres uddannelse.
- Mange forskellige emner: nogen meget nemt andre mere udfordrende.

Cell Metabolism

Resource

Plasticity of Epididymal Adipose Tissue in Response to Diet-Induced Obesity at Single-Nucleus Resolution

Graphical Abstract



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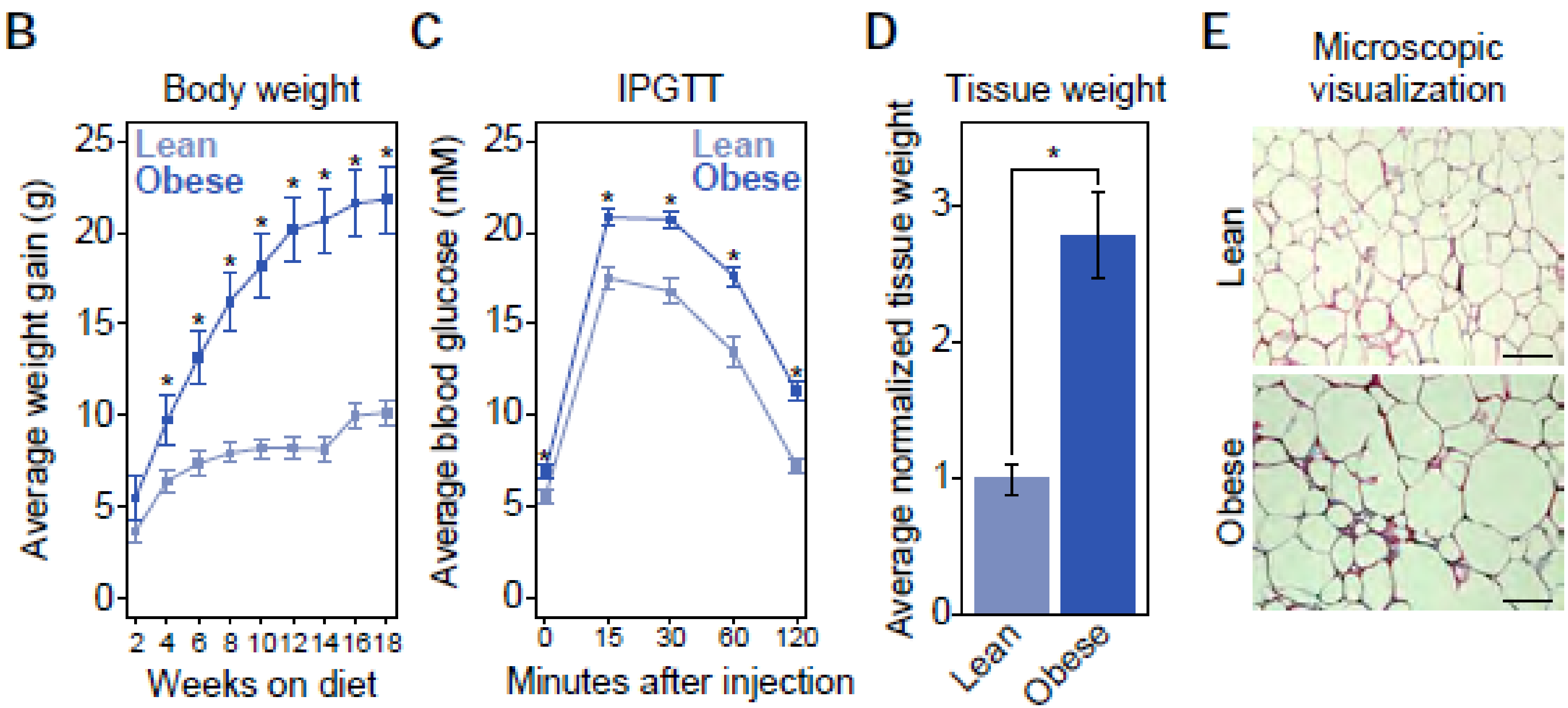
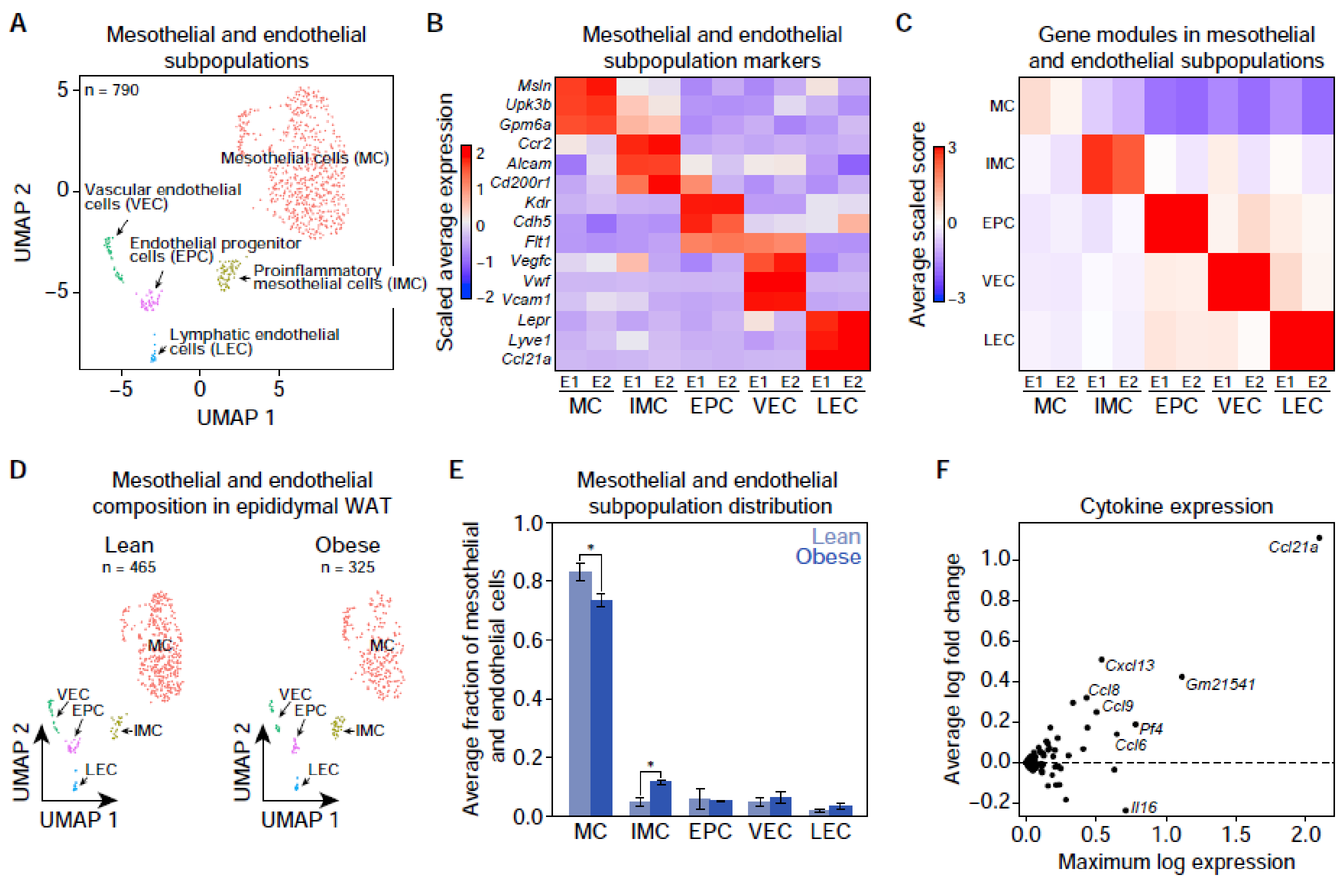
In Brief

Using single-nucleus RNA-seq, Sárvári et al. show that diet-induced obesity leads to major transcriptional changes in all cell types in the epididymal adipose tissue. They show that *in vivo* adipocyte differentiation is affected by obesity and demonstrate the existence of three adipocyte subpopulations, one of which is lost in obesity.


B
Body weight
Lean
Obese
Average weight gain (g)
0
5
10
15
20
25
2 4 6 8 10 12 14 16 18
Weeks on diet
C
IPGTT
Lean
Obese
Average blood glucose (mM)
0
5
10
15
20
25
0 15 30 60 120
Minutes after injection
D
Tissue weight
Average normalized tissue weight
0
1
2
3
Lean
Obese
E
Microscopic visualization
Lean
Obese


A
Mesothelial and endothelial subpopulations
n = 790
Mesothelial cells (MC)
Vascular endothelial cells (VEC)
Endothelial progenitor cells (EPC)
Proinflammatory mesothelial cells (IMC)
Lymphatic endothelial cells (LEC)
UMAP 2
UMAP 1
B
Mesothelial and endothelial subpopulation markers
Msln
Upk3b
Gpm6a
Ccr2
Alcam
Cd200r1
Kdr
Cdh5
Flt1
Vegfc
Vwf
Vcam1
Lepr
Lyve1
Ccl21a
Scaled average expression
E1 E2
MC IMC EPC VEC LEC
C
Gene modules in mesothelial and endothelial subpopulations
Average scaled score
MC
IMC
EPC
VEC
LEC
D
Mesothelial and endothelial composition in epididymal WAT
Lean
n = 465
Obese
n = 325
E
Mesothelial and endothelial subpopulation distribution
Lean
Obese
Average fraction of mesothelial and endothelial cells
F
Cytokine expression
Ccl21a
Cxcl13
Gm21541
Ccl8
Ccl9
Pf4
Ccl6
Il16
Average log fold change
Maximum log expression

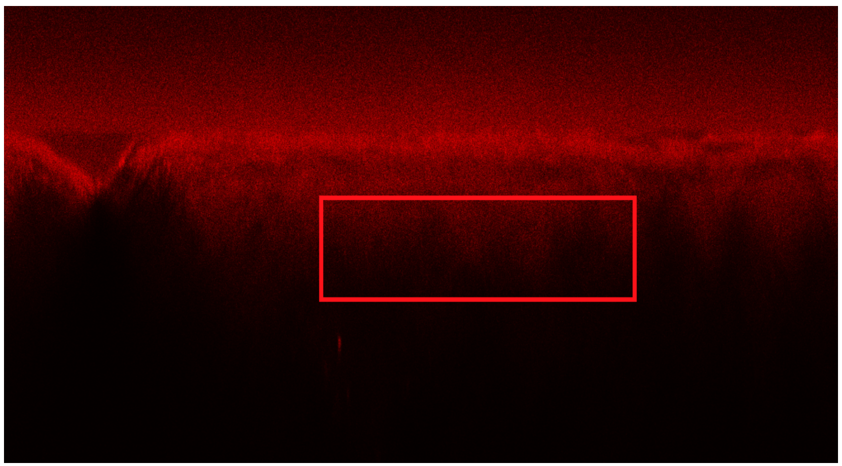
Data visualisering

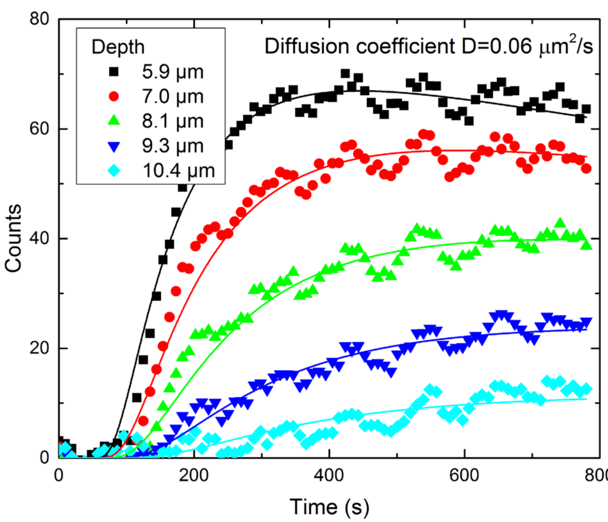
AG17691 | 0.042733989

		epi.vs.derm		epi.vs.a375		epi.vs.sk_mel		epi.vs.ft		epi.vs.sk_mel_epi		epi.vs.sph_a375		epi.vs.sph_
Transcript	annotation	log2FC	p_adjust	log2FC	p_adjust	log2FC	p_adjust	log2FC	p_adjust	log2FC	p_adjust	log2FC	p_adjust	log2FC
NM_001456	FLNA|ABP-280|ABPX|CSB	0.58921196	0.10962767	0.31251343	0.32151865	-0.52623445	0.10834491	0.69965596	0.06030511	-0.55687138	0.1909472	-0.27371765	0.33670851	-0.090743
NM_201380	PLEC|EBS1|EBSMD|EBSNE	-0.07310948	0.86324259	0.35826042	0.17723474	-0.38559429	0.20285444	0.28726553	0.50515563	-0.55780897	0.07213527	-0.01183482	0.95255894	0.3167832
NM_001961	EEF2|EEF-2|EF-2|EF2|SCA	-0.15425131	0.67954951	0.10080635	0.74907898	0.25382608	0.39729394	0.31418909	0.36713837	-0.00669064	0.99026062	-0.13911671	0.61414632	-0.41849888
NM_001620	AHNAK|AHNAKRS|PM227	0.06059442	0.87728319	0.01508445	0.96470577	-0.27222901	0.42698693	-0.22843189	0.57466546	-0.22338423	0.62943464	-1.96316403	**1.36E-18**	-1.70933717
NM_212482	FN1|CIG|ED-B|FINC|FN|F	3.481258	**1.65E-24**	3.26375829	**2.92E-35**	3.33898238	**3.92E-37**	3.43235645	**3.73E-29**	2.85648962	**4.90E-17**	0.46199353	0.10710672	-0.33115316
NM_002473	MYH9|BDPLT6|DFNA17|EI	-0.82499807	**0.00635195**	-0.55185218	**0.0347997**	-0.75415619	**0.00530423**	0.16973352	0.66357462	-0.85151383	**0.00484467**	-2.19867131	**2.58E-27**	-2.09884234
NM_001101	ACTB|BRWS1|PS1TP5BP1|	-1.03095662	**0.00083716**	-1.01159699	**2.16E-05**	-1.01387454	**2.26E-05**	-0.18388955	0.61781852	-1.15124951	**8.90E-05**	-1.62899926	**3.17E-12**	-1.95105347
NR_002819	MALAT1|HCN|LINC00047|	-1.24440963	**0.00283003**	-1.19717692	**0.00015086**	-0.31566781	0.42472554	-1.39428042	0.16000025	-0.91330158	**0.02924663**	-1.18939401	**0.00020129**	-0.95357976
NM_001614	ACTG1|ACT|ACTG|DFNA2	-0.87633673	**0.00678357**	-0.61939194	**0.01806608**	-0.57642263	**0.03262907**	0.0497931	0.91421652	-0.73842319	**0.0278376**	-1.13170037	**3.98E-06**	-1.55073059
NM_002654	PKM|CTHBP|HEL-S-30|OIF	-0.86560246	**0.00872766**	-0.05949325	0.85971991	-0.22050013	0.49825687	0.30253245	0.39812994	-0.53648851	0.15306615	-0.42999787	0.10481768	-0.57705535
NM_005063	SCD|FADS5|MSTP008|SCD	-1.11431088	**3.45E-05**	-0.36347282	0.05981058	-0.53181786	**0.00385002**	-0.28811047	0.31799004	-0.68997704	**0.00214592**	-1.70810618	**2.17E-34**	-2.20184091
NM_014762	DHCR24|DCE|Nbla03646|S	-1.5965058	**2.71E-11**	-0.67839304	**0.00038442**	-0.64496438	**0.00048917**	-0.26970294	0.34737999	-0.15639226	0.63042304	-0.71459807	**1.52E-06**	-0.71322052
NM_002568	PABPC1|PAB1|PABP|PAB	-0.96021975	**0.00052714**	-0.43234187	0.08257409	0.00496745	0.98701824	-0.22299403	0.45852778	-0.3842191	0.15653428	-0.78568571	**6.44E-05**	-1.3526768
NR_145459	TALAM1|-|-|-|ncRNA	-1.20764404	**0.00413853**	-1.16188791	**0.00027617**	-0.2969723	0.46206266	-1.37507865	0.16867079	-0.89374602	**0.03630019**	-1.15750185	**0.00035133**	-0.91588168
NM_002160	TNC|150-225|DFNA56|GM	2.63441432	**1.53E-13**	1.37821172	**8.54E-06**	1.36543825	**1.47E-05**	1.9310747	**1.73E-08**	0.64316557	0.18362998	2.0140446	**7.38E-13**	2.29167731
NM_022833	FAM129B|C9orf88|MEG-3	-0.74491021	**0.01732943**	-0.36288191	0.1544667	-0.31407447	0.26359551	-0.02583226	0.95455465	-0.47552164	0.16810234	-0.7043883	**0.0013669**	-0.64595191
NM_000484	APP|AAA|ABETA|ABPP|A	-0.55398382	0.05984762	-0.47003234	**0.02965794**	-0.38980788	0.08921669	-0.01279339	0.97430606	-0.32176078	0.31286958	-0.83373954	**4.01E-06**	-0.92065015
NM_170707	LMNA|CDCD1|CDDC|CMD	-0.40429365	0.26709905	-0.23004867	0.45682282	-0.29497922	0.36573071	0.18722334	0.64785212	-0.4105158	0.31859292	-0.54048596	**0.04723619**	-0.94139101
NM_147782	CTSB|APPS|CPSB|-|8p23.	-0.63413917	**0.01947499**	-0.8697254	**7.15E-06**	-0.22230285	0.32295353	-0.05823308	0.87483831	-0.51409724	**0.04811676**	-1.15865103	**1.28E-12**	-0.97232663
NM_005336	HDLBP|HBP|PRO2900|VGI	0.25650374	0.365858	-0.06056658	0.799615	-0.07788875	0.76803104	0.3266535	0.27203611	-0.20828049	0.52851523	-0.65445588	**1.02E-05**	-0.61700897
NM_000421	KRT10|BCIE|BIE|CK10|EHI	-3.13820066	**3.21E-07**	-1.03303235	**0.00021378**	-0.79026427	**0.00193596**	-0.19889726	0.59495463	-0.41770205	0.26078251	-5.73269521	**1.20E-144**	-5.50890978
NM_004104	FASN|FAS|OA-519|SDR27	0.16954341	0.6323711	1.06659688	**3.91E-09**	0.1940028	0.33855299	0.40753153	0.27858102	0.35541998	0.17904628	0.69633169	**5.31E-06**	0.7523758
NM_001418	EIF4G2|AAG1|DAP5|NAT1	-0.68917992	**0.01224337**	-0.70690465	**0.00039477**	-0.39849813	**0.04266541**	-0.30209163	0.31008817	-0.51176311	**0.03871046**	-1.3583528	**5.71E-23**	-1.86746404
NM_0012911	EIF4G1|EIF-4G1|EIF4F|EIF4	-0.62599227	**0.02619945**	-0.57384358	**0.01077187**	-0.52770655	**0.02240952**	-0.03267743	0.94095135	-0.3845997	0.21798167	-0.98698949	**1.97E-08**	-1.08006435
NM_004924	ACTN4|ACTININ-4|FSGS|F	-0.98857066	**0.00094647**	-0.45109661	0.0675203	-0.68812061	**0.00579045**	-0.01765814	0.96971155	-0.71113461	**0.02443112**	-0.46248494	**0.04503417**	-0.56502391
NM_004415	DSP|DCWHKTA|DP|-|6p24	-3.38609382	**0.00451919**	-0.65264254	**0.00488513**	-0.51137507	**0.03006599**	-0.37408374	0.19179683	-0.49577597	0.08828619	-5.89611778	**8.26E-253**	-5.38875149
NM_000918	P4HB|CLCRP1|DSI|ERBA2L	-0.4569514	0.1175118	-0.36275439	0.11459978	-0.29887796	0.21002106	0.07304192	0.84749658	-0.32597037	0.30306058	-0.93717262	**2.66E-07**	-0.84913489
NM_014631	SH3PXD2A|FISH|SH3MD1|	0.81235697	**0.01050942**	-0.15374839	0.60884762	-0.13213909	0.72808992	-0.40102562	0.25102059	-0.12178527	0.81208306	-0.53907209	**0.01956228**	-0.40802525
NM_000177	GSN|ADF|AGEL|-|9q33.2|	-0.43498325	0.20953143	-0.27321209	0.33963774	-0.11346565	0.74862455	0.0734768	0.87223436	-0.44826363	0.24207085	-0.69272822	**0.00755127**	-0.72838297
NM_000701	ATP1A1|-|-|1p13.1|prote	-1.17776172	**0.00020068**	0.42406145	0.1012868	0.75325609	**0.00358857**	0.39103847	0.23754886	0.523417	0.14759896	0.68188509	**0.00331875**	0.53319927
NM_002016	FLG|ATOD2|-|1q21.3|prot	-3.80882593	**5.69E-10**	-1.34272194	**0.04539129**	-0.43621047	0.32222507	-0.78268641	0.3423076	-0.07877552	0.91845735	-5.29661105	**6.76E-47**	-4.22230489
NM_145690	YWHAZ|14-3-3-zeta|HEL-S	-1.05311543	**0.00110252**	-1.06610949	**2.05E-06**	-0.80026512	**0.00012108**	-0.44033702	0.18731336	-0.87807429	**0.00093036**	-1.74411606	**1.41E-21**	-2.34339143
NR_027451	NORAD|LINC00657|-|20q1	-0.05801054	0.86723098	0.08100792	0.74736123	0.3407029	0.12822574	-0.18668563	0.58627683	0.17441375	0.60345383	0.30832837	0.06365292	-0.08247627
NM_0013052	AGRN|AGRIN|CMS8|CMSI	0.20324203	0.58502824	1.65321027	**4.55E-16**	0.96642185	**3.95E-06**	1.26754101	**1.11E-05**	0.86250645	**0.00264354**	0.92073368	**6.21E-07**	1.28309446
NM_0012428	CSDE1|D1S155E|UNR|-|1p	-0.56739338	0.06313381	-0.34599081	0.09047017	-0.07817066	0.73607589	-0.31382867	0.25941456	-0.2780605	0.28575677	-0.37537604	**0.00602215**	-0.80299114
NM_003246	THBS1|THBS|THBS-1|TSP|	0.69144868	**0.01149898**	-0.00597148	0.98272625	0.19101435	0.39220761	0.61036479	**0.02727718**	0.23856829	0.39186107	-2.39317038	**3.40E-20**	-3.07913105
NM_203330	CD59|16.3A5|1F5|EJ16|EJ	-1.25462691	**4.56E-07**	-0.95435109	**5.62E-09**	-0.46794048	**0.00888467**	-0.41486857	0.11240361	-1.17028909	**5.76E-09**	-0.9330295	**1.23E-13**	-1.20859529
NM_003118	SPARC|BM-40|OI17|ON|-	4.89921456	**8.55E-84**	4.3211658	**1.52E-121**	4.21510141	**3.22E-111**	5.02993796	**9.77E-121**	4.5061181	**7.92E-101**	4.1971228	**1.67E-121**	3.83421156
NM_005348	HSP90AA1|EL52|HEL-S-65p	-1.11825075	**0.00060449**	-0.46807477	0.05912348	-0.46868991	0.06957655	-0.09675493	0.82239572	-0.65355085	**0.04217297**	-0.83096431	**0.00012351**	-1.31506603
NM_004995	MMP14|MMP-14|MMP-X1	0.64987278	**0.0398817**	0.33344993	0.20316019	0.31080969	0.2691587	0.74778616	**0.00814349**	0.44456743	0.20840391	-1.53114781	**5.28E-13**	-1.6392745
NM_006597	HSPA8|HEL-33|HEL-S-72p|	-0.90243542	**0.01210693**	-0.62518543	**0.02996501**	-0.55867351	0.06809746	-0.03291725	0.94830567	-0.74162375	**0.04899193**	-0.57982088	**0.03809811**	-0.92104156
NM_021960	MCL1|BCL2L3|EAT|MCL1-E	-1.14016714	**2.76E-05**	-0.34453829	0.06086982	-0.0647225	0.8062385	-0.15710131	0.6207984	-0.26273936	0.33714616	0.24910007	0.12929796	-0.21063003
NM_0010774	GNAS|AHO|C20orf45|GNA	-0.63840548	**0.02970178**	0.29778207	0.22602046	-0.00544327	0.98748697	0.05240432	0.89958574	-0.53637143	0.07958964	-0.30118751	0.1793881	-0.55386867
NM_000610	CD44|CDW44|CSPG8|ECM	-0.86893147	**0.00212414**	-0.19805083	0.28855549	0.11123388	0.61260164	-0.09832506	0.77575203	-0.1420132	0.65369068	-0.9085429	**1.90E-11**	-1.55992095
NM_014045	LRP10|LRP-10|LRP9|MST0	-1.78712294	**2.67E-14**	-1.27806892	**1.70E-14**	-0.79485573	**1.04E-05**	-0.78491534	**0.00068276**	-0.73312051	**0.00070794**	-1.99642088	**5.54E-57**	-1.60620046
NM_0012014	ENO1|ENO1L1|HEL-S-17|M	-0.84022397	**0.01310905**	0.18442151	0.56309596	-0.03627185	0.93200387	0.37341423	0.28812787	-0.43309129	0.28501713	-0.37282946	0.18168167	-0.6763564
NM_012401	PLXNB2|MM1|Nbla00445|	0.35584875	0.24216329	0.21326947	0.3488715	0.11773641	0.68432613	0.43755856	0.23371351	0.13582217	0.73040645	-0.84211623	**8.10E-08**	-0.28211881
NM_003750	EIF3A|EIF3|EIF3S10|P167|	-0.21722131	0.51938906	-0.02051102	0.93679205	0.23528649	0.36198214	-0.05277648	0.89410414	0.0114943	0.97939656	0.05377783	0.78145873	-0.32314304
NM_003217	TMBIM6|BAXI1|BI-1|TEGT	-0.93657199	**0.00028337**	-0.55509038	**0.00567124**	-0.34181705	0.0878972	-0.15995254	0.62227955	-0.42704329	0.10117806	-0.83670511	**2.08E-08**	-1.02015126
NM_0012719	HSP90AB1|D6S182|HSP84	-0.42063432	0.28667772	0.18041124	0.58192394	0.09551201	0.8110864	0.34113219	0.3622012	-0.26231061	0.58522394	0.17585587	0.55807963	-0.19775581
NM_178014	TUBB|CDCBM6|CSCSC1|M	0.05571883	0.88016504	-0.03049732	0.92104708	-0.55254073	**0.02864046**	0.51150579	0.07758594	-0.7851343	**0.00936073**	-0.57654266	**0.00963718**	-1.01074921
NM_001642	APLP2|APLP-2|APPH|APP	-0.1192064	0.71209426	-0.69391831	**0.00031735**	-0.3265763	0.13402166	-0.24528841	0.41807663	-0.49215399	0.06881619	-1.12948991	**4.49E-12**	-1.06724347
NM_0011300	SHC1|SHC|SHCA|-|1q21.3	-0.92538896	**7.58E-05**	-0.48605761	**0.00673789**	-0.39575627	**0.03597792**	-0.55328937	**0.01828552**	-0.84741773	**8.42E-05**	-0.73237471	**3.27E-08**	-0.74296959
NM_012232	CAVIN1|CAVIN|CGL4|FKS	0.59998693	0.05102366	-0.04559367	0.87778138	-0.41009651	0.11606847	0.61128644	**0.02586152**	-0.19313047	0.63409592	-0.67375373	**0.00110876**	-0.72983599

Fit data: Penetration of D_2O into skin

Add D_2O and record intensity as function of depth and time.





Fit to simple diffusion model $\Delta C = \frac{a}{\sqrt{t}}\exp(-x^2/4Dt)$

$D=0.06\mu m^2/s$

Formålet ved kurset

- Lære jer nødvendig matematik og introducere data science kompetencer som i vil have bruge for i løbet af jeres uddannelse.
- Mange forskellige emner: nogen meget nemt andre mere udfordrende.

Kursets opbygning

2 semester

1. semester

- 16 timers forelæsninger
- 20 timers e-timer
- 16 timers computer lab

Eksamen

-Digitale eksamen

Forelæsninger:

Underviser

- Jonathan Brewer brewer@bmb.sdu.dk
- Kristian Debrabant
- Veit Schwammle
- Richard Röttger

Plans

Itslearning

Mange forskellige emner

Lecture 1: Calculations and analysis of mathematical functions relevant... LECTURES FALL 2024

Lecture: Chapter 2 and 4 in "Mathematics for the Life Sciences". Learning goals: 1) Understand the importance of data scienc in life science. a. Why is data science important for your studies? 2) Review exponential and logarithmic functions. a. Why are

4. sep. 10.00–12.00 2 resources

Lecture 2: Basic programming / Introduction to R: Veit Schwämmle LECTURES FALL 2024

Introduction to R

11. sep. 10.00–12.00 2 resources

Lecture 3: Data structures: RR LECTURES FALL 2024

Introduction to data structures in R

23. sep. 17.00–19.00 1 resource

Lecture 4: Applied statistics. Veit Schwämmle LECTURES FALL 2024

Applied statistics. Chapters 1, 2, 10.1-10.2 and 25

2. okt. 10.00–12.00 1 resource

Lecture 5: Data modeling/regression: KD LECTURES FALL 2024

Data modeling/regression. Chapters 3-4 in "Mathematics for the Life Sciences". Learning goals: Understand the concept of least squares regression Be able to perform linear least squares regression in R Understand the concepts of linear

8. okt. 14.00–16.00 1 resource

Lecture 6: Multiple Linear Regression (MLR) Interpret coefficients of a... LECTURES FALL 2024

Multiple Linear Regression (MLR) Interpret coefficients of an MLR: RR

24. okt. 16.00–18.00 1 resource

Lecture 7: Linear algebra. KD LECTURES FALL 2024

Linear algebra including vectors, matrices, solution linear systems of equations, determinants, eigenvalues and eigenvectors. Chapters 6-7 in "Mathematics for the Life Sciences" Learning goals: 1. Be able to calculate 2x2 determinants 2. Learn how to

6. nov. 10.00–12.00 2 resources

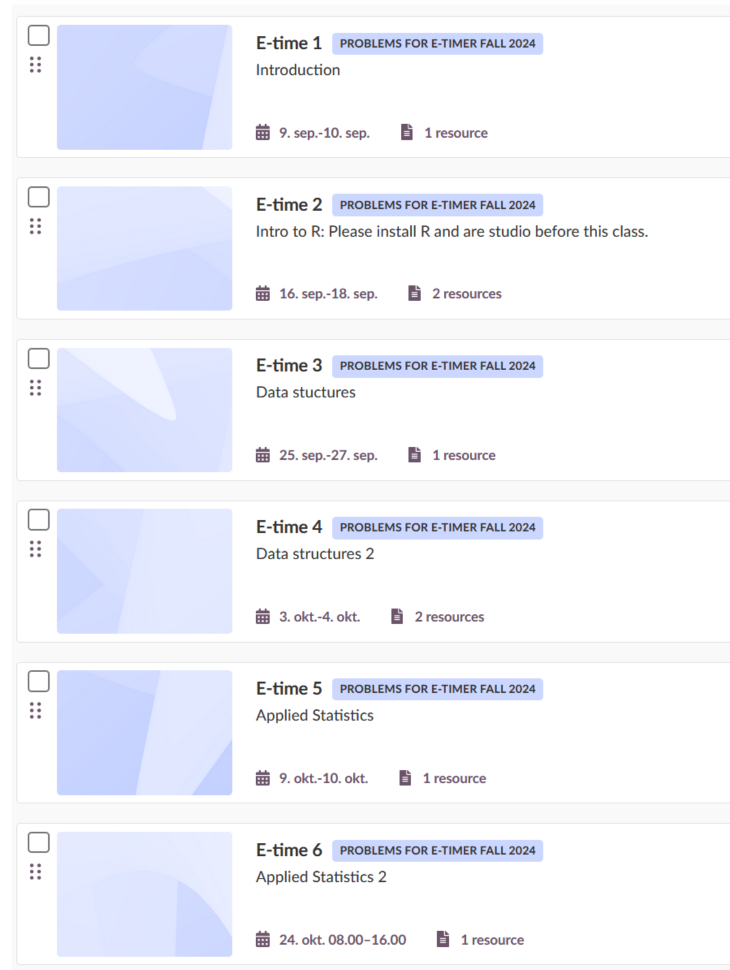
Lecture 8: Biological applications of derivative. JB/KD LECTURES FALL 2024

Rates of change. Biological applications of derivatives max and mean. Chapters 17 and 20 in Mathematics for the Life Sciences. Learning goals: Understand average rates of change. Understand applications / the meaning of rates of change in

13. nov. 10.00–12.00 2 resources

E-timer

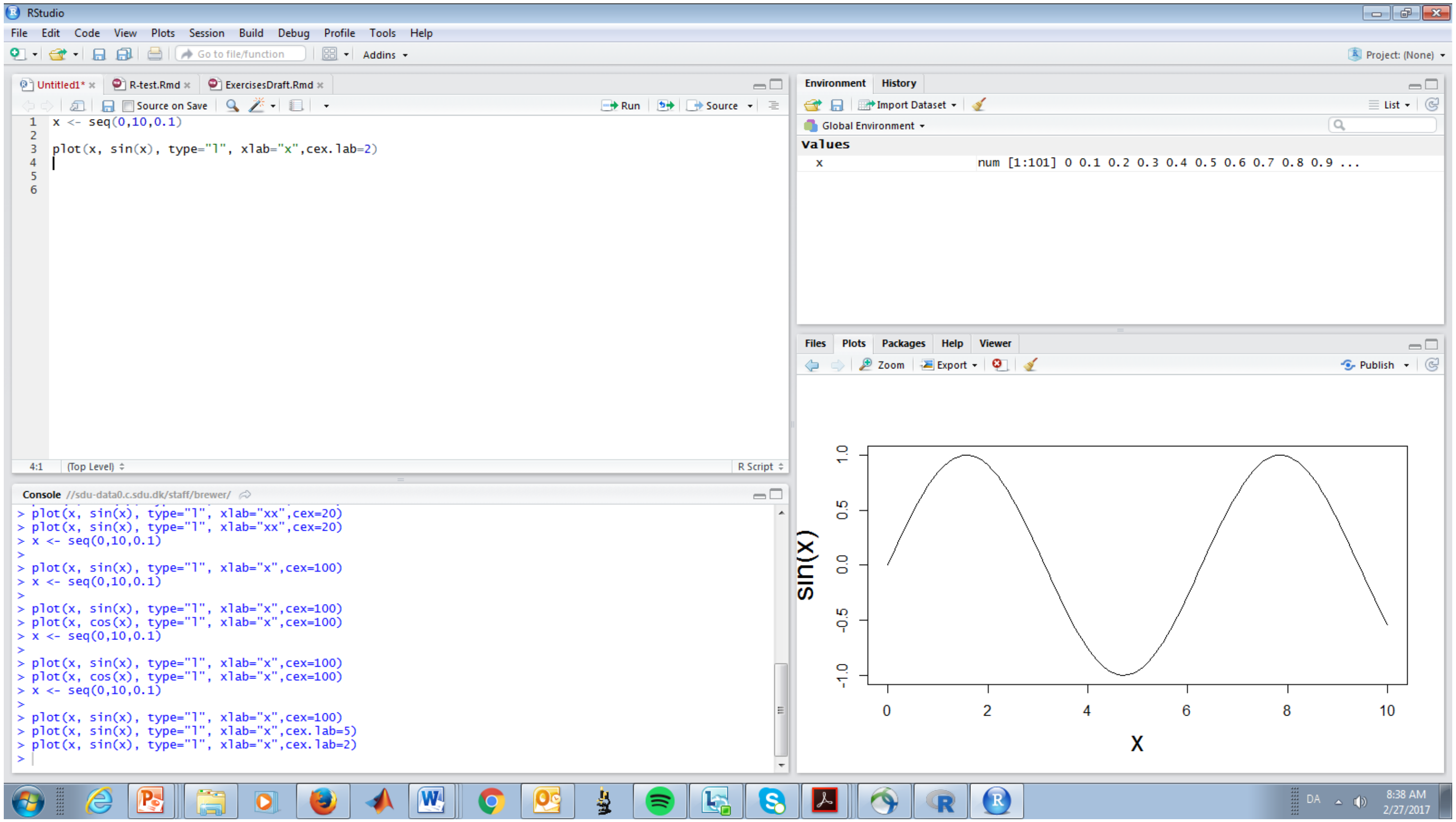
- Undervisning med fysisk fremmøde
- Prøv at løs opgaverne selv
- Diskuter opgaven med en kammerat eller i en gruppe
- Prøv at løse opgaven igen
- Deltage aktivt i E-timerne



LAB

- 4 computer lab I løbet af kurset.
- Der er møde pligt!!
- Vi vil bruge datacamp.
- I vil arbejde sammen jeres grupper.
- Man skal have bestået de obligatorisk moduluer i datacamp. Det svare til 1 rapport aflevering.
- Der skal skrives en Rapporter I løbet af kurset. Bestået/ikke bestået.
- Rapporter skal afleveres igennem Itslearning.

Hvorfor R ?



Der forventes at i lære at arebjde i R. Det kræver arbejde!!!

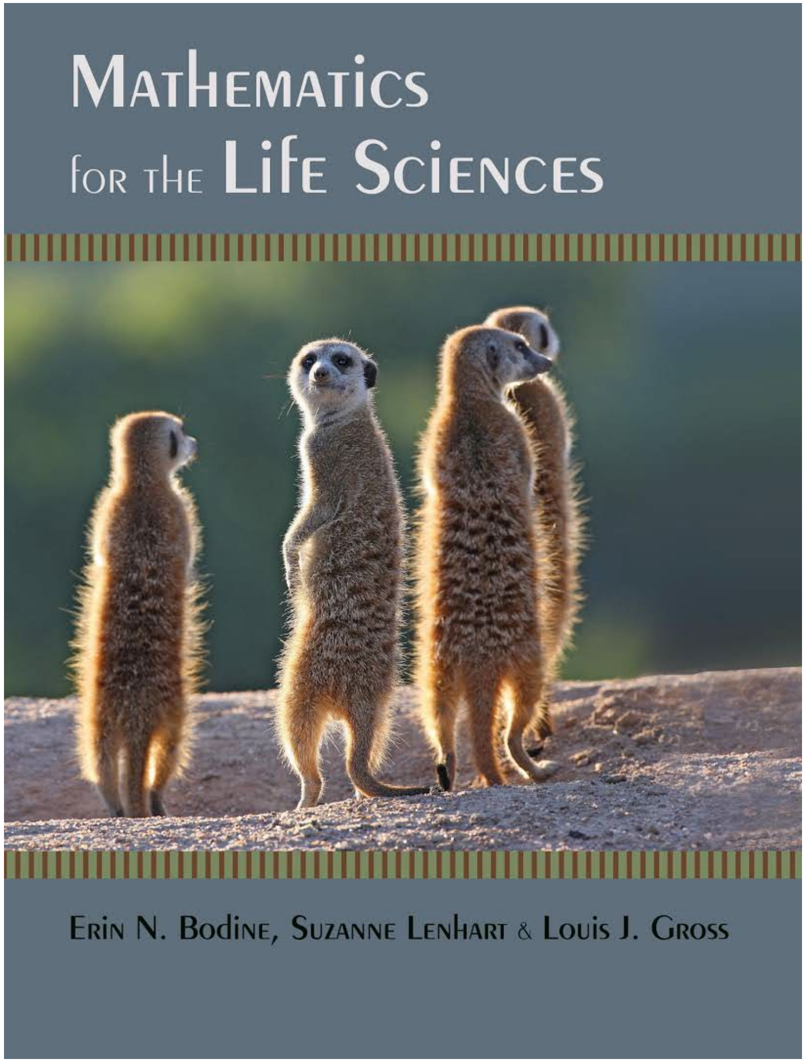
How to learn R?

- Teach me how to make a graph in R

VS

- Please create an R script to plot enzyme activity using real data. The script should use the Michaelis-Menten equation to find and show Km and Vmax values on the graph.

Bogen



Pensum står beskrevet under resources/Course materials og i de forskellige plans.

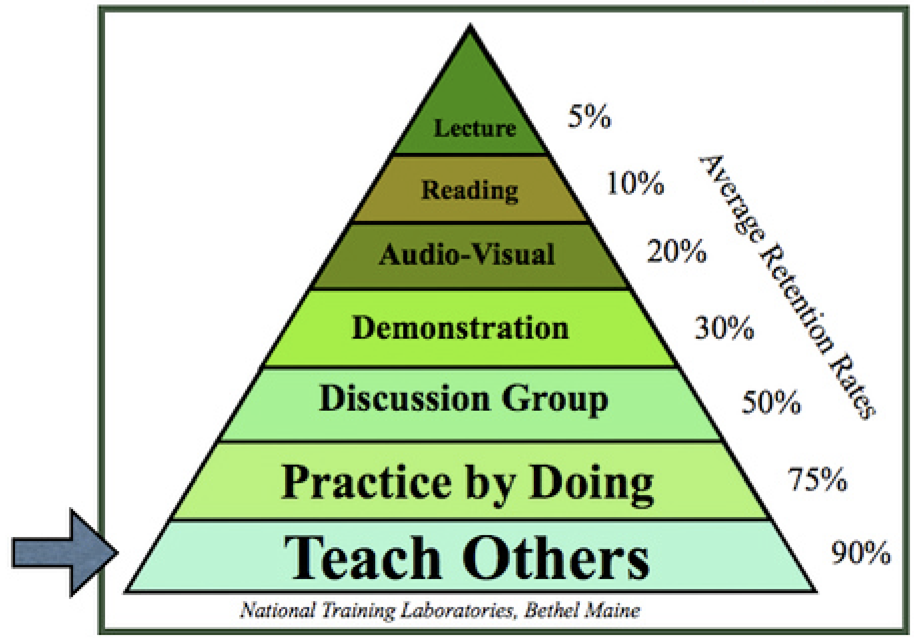
Formålet ved kurset

- Lære jer nødvendig matematik og introducere data science kompetencer som i vil have bruge for i løbet af jeres uddannelse.
- Mange forskellige emner: nogen meget nemt andre mere udfordrende.
- Strategi for kurset.
  - Se forelæsninger aktivt , tage feks. noter undervejs
  - regne opgaverne, gerne med andre.

Så vil i komme fint igennem kurset☺

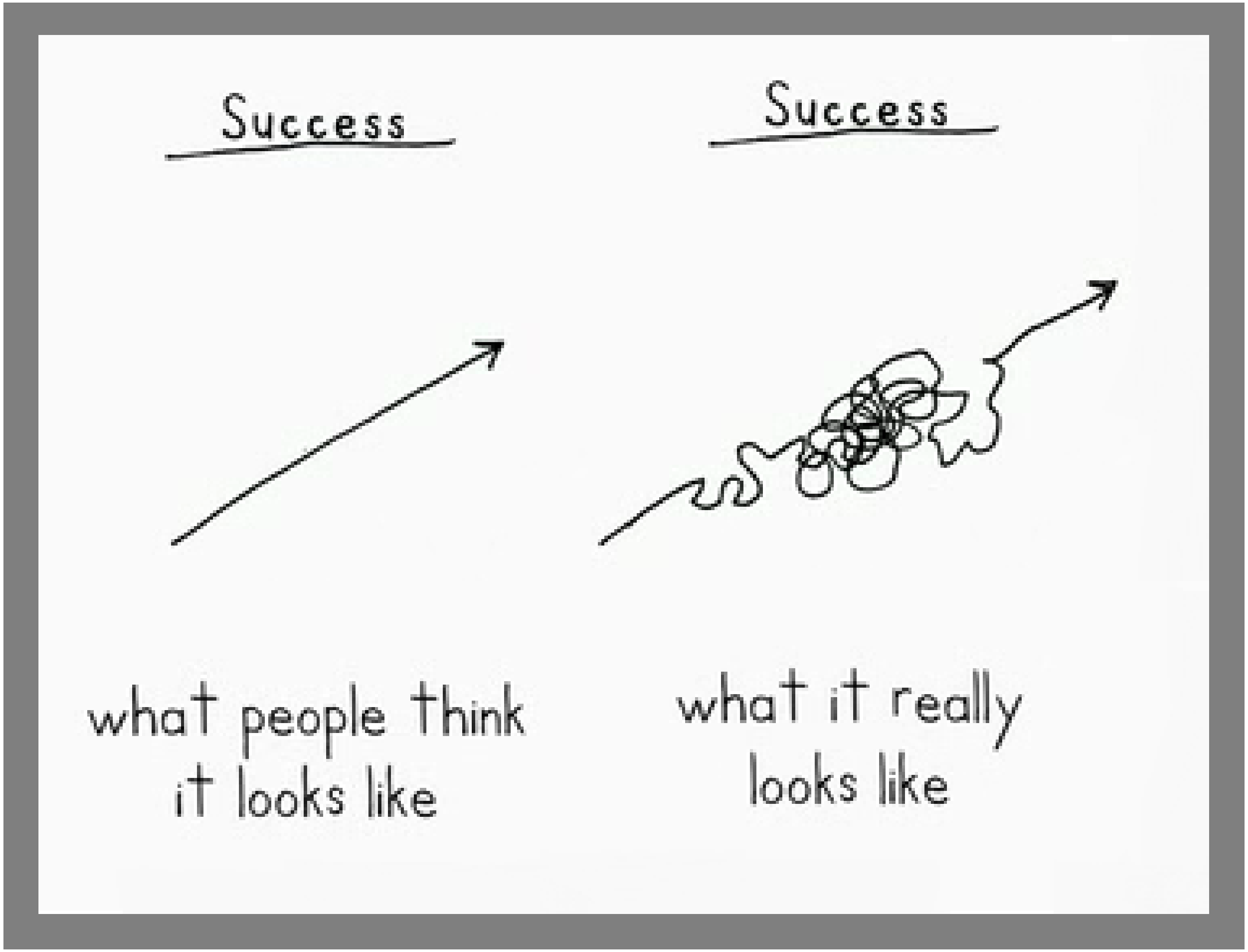
The Learning Pyramid

- 5% retention from listening to lecture
- 10% when you read for information
- 20% with use of audio-visual
- 30% comes with a demonstration added
- 50% when you discuss facts in groups
- 75% when you practice by doing
- 90% when you teach others as you learn
- (From the National Training Institute in 1999)



Is learning something we all just can do? Or are there some strategies better than others?

"Talent" er overvurderet: Bevidst øvelse er vejen til ekspertise


Success
Success
what people think it looks like
what it really looks like

BMB547

Molekylær Data Science

Introduktion

Emner for i dag

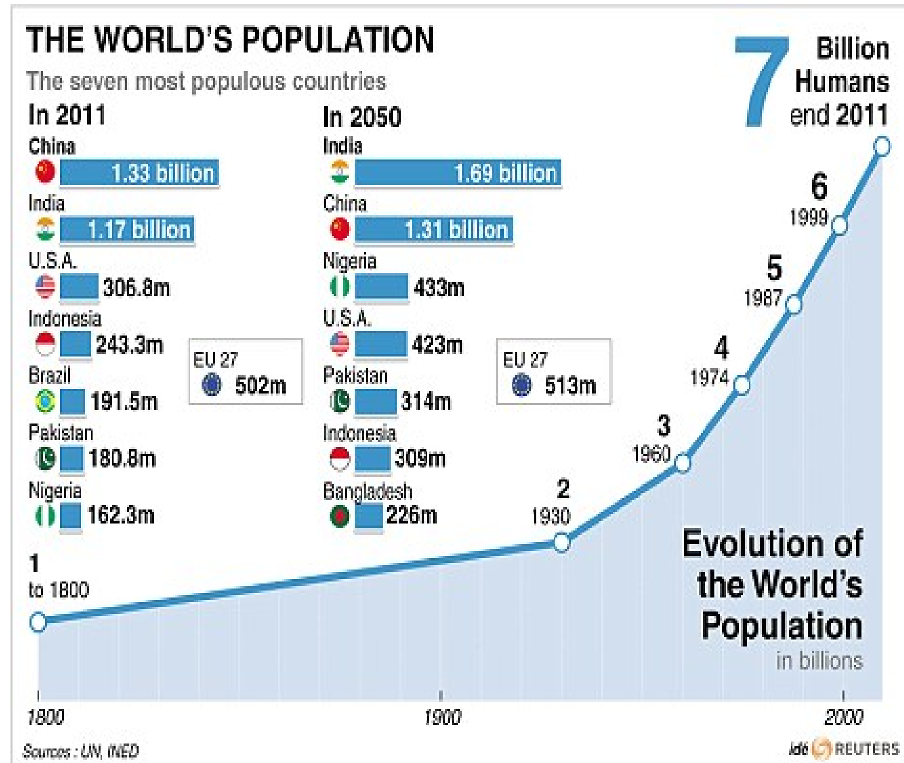
Eksponentielle funktioner

Enheder / Koncentrationer

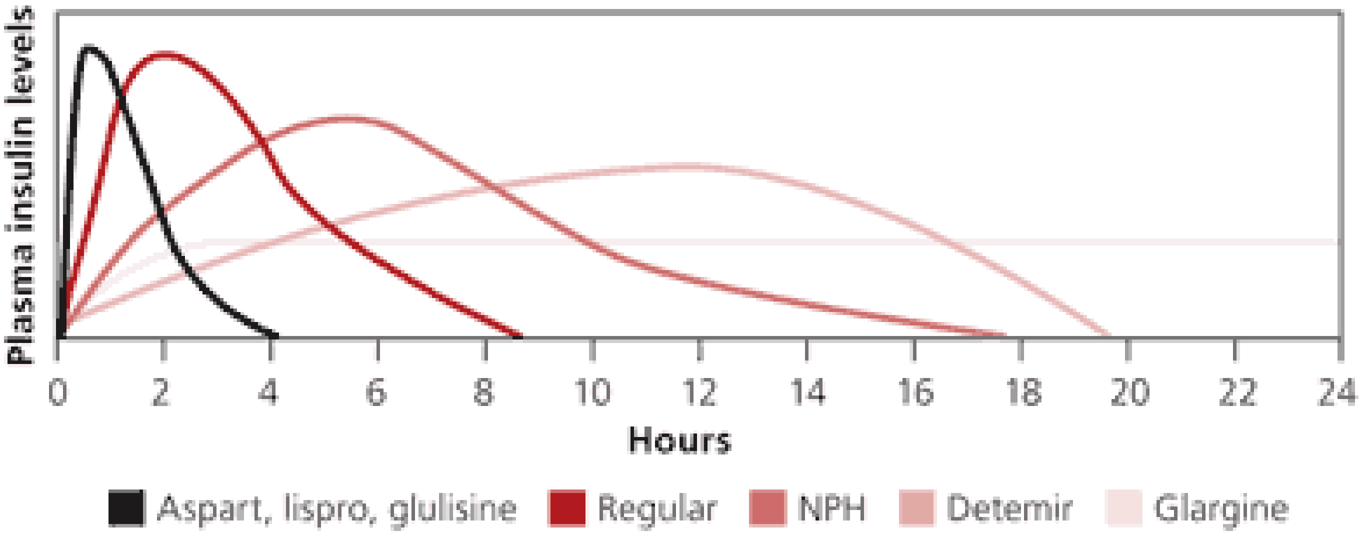
Datarepræsentation

Eksempler på eksponentielle modeller

- Penge/Investeringer
- Halveringstid af lægemiddel i kroppen
- Farmakokinetik
- Bakterievækst
- Befolkningsvækst
- Beer-Lamberts lov
- Gibbs fri energi
- pH



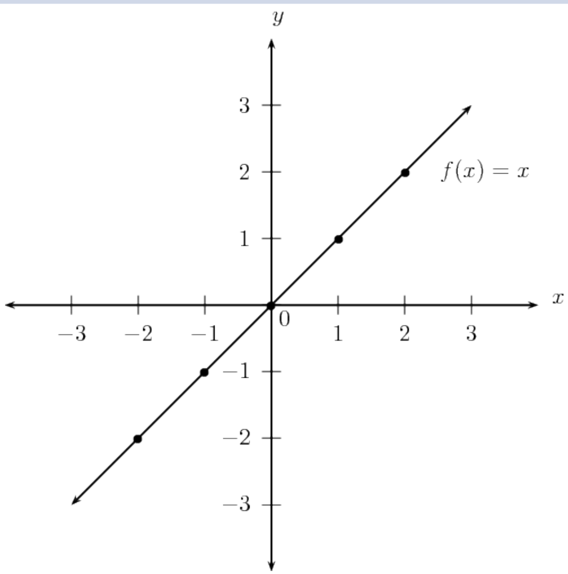
Farmakokinetik

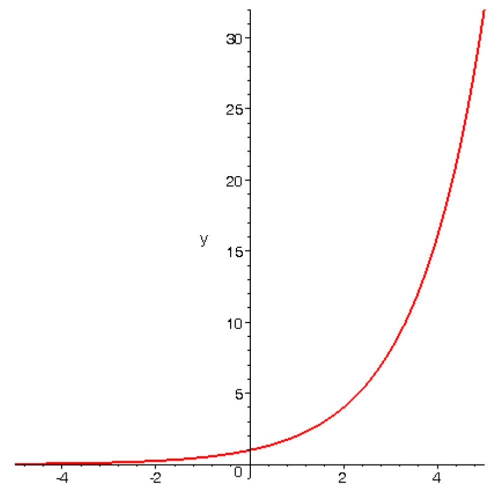


Begyndelse af virkning, toppunkt og
varighed af eksogene insulinpræparater

Lad os sammenligne lineære funktioner og eksponentielle funktioner

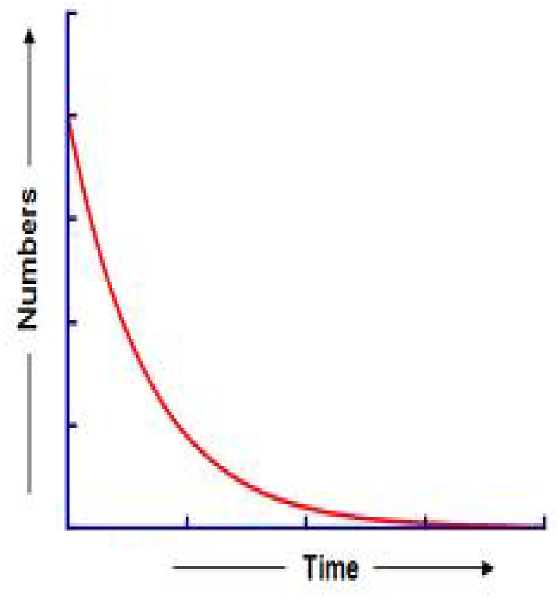
Linear Function	**Exponential Function**
• Change at a constant rate • Rate of change (slope) is a constant	• Change at a changing rate • Change at a constant <u>percent</u> rate



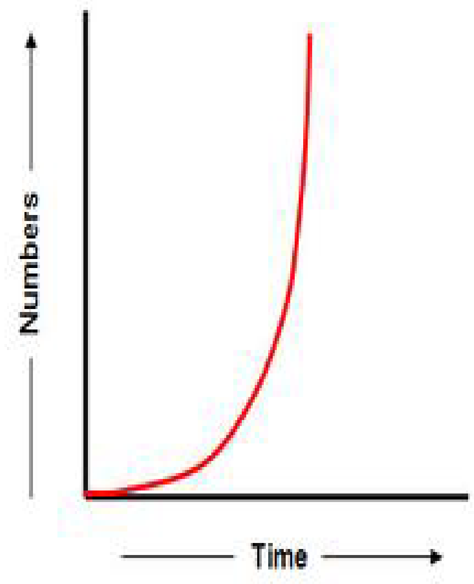


Hvordan kan du afgøre, om en eksponentiel funktion modellerer vækst eller henfald, blot ved at se på dens graf?

Graph 1

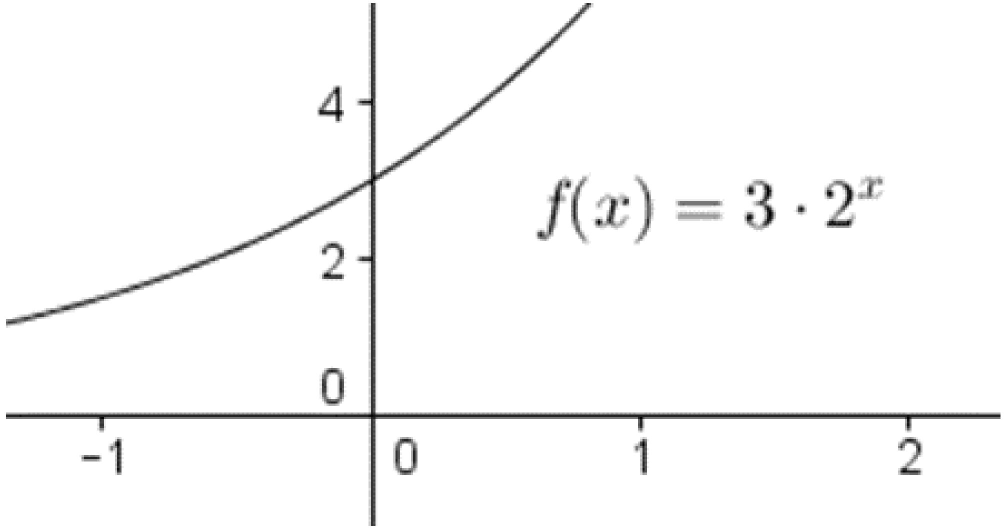


Graph 2



Hvordan kan vi definere eksponentielle funktioner symbolsk?

- $f(x) = ab^x$
- Bemærk, at variablen er i eksponenten?
- **Basen er b**, og **a er koefficienten** og er > 0.
- Denne koefficient er også begyndelsesværdien/y-aksens skæring (når x = 0).



Sammenligning af eksponentiel vækst/henfald i forhold til deres ligninger

Eksponentiel vækst

$f(x) = a \cdot b^x$ for $b > 1$

Eksempel:

$f(x) = 2^x$

Eksponentiel henfald

$f(x) = a \cdot b^x$ for $0 < b < 1$

Eksempel :

$$f(x) = \left(\frac{1}{2}\right)^x$$

Laws of Exponents

1. $$a^x a^y = a^{x+y}$$

2. $$\frac{a^x}{a^y} = a^{x-y}$$

3. $$\left(a^x\right)^y = a^{xy}$$

4. $$\left(ab\right)^x = a^x b^x$$

5. $$a^0 = 1$$

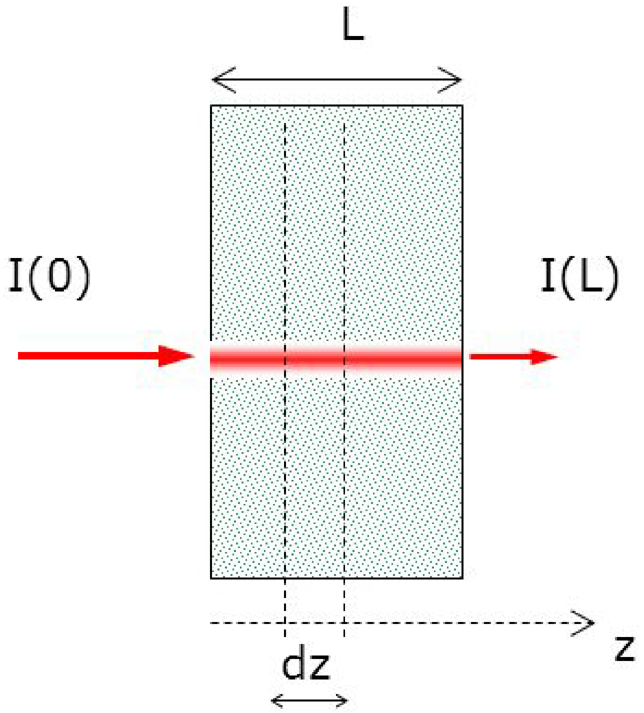
6. $$\frac{1}{a^x} = a^{-x}$$

What's with exp(x)=e^x where e=2.71828

1. **Eksponentiel vækst i biologiske processer**
   - **e** modellerer kontinuerlig biologisk vækst, som bakterievækst, populationsudvikling og sygdomsspredning.
2. **Unik aflednings-egenskab**
   - Funktionen e^x har den samme ændringshastighed som sin værdi, hvilket forenkler beregninger i farmakokinetik og biologiske reaktioner.
3. **Kontinuerlig sammensat vækst**
   - **e** repræsenterer kontinuerlig ændring, hvilket er afgørende for at forstå processer som stofmetabolisme og næringsstofoptagelse.

Lambert Beer

Light attenuation in a clear medium



$$I(z)=I_0*a^z$$

I = light intensity [W cm^{-2}]

μ_a = absorption coefficient [cm^{-1}]

L = source-detector distance = pathlength [cm]

Lambert Beer

$I(x)=I_0*a^x$

If the flux radiant density is ¼ at 1 meters depth what is a?

1

0%

1/4

0%

ln(1/4)

0%

1/2

0%

4

0%

Lambert Beer

$I(x)=I_0*a^x$

If the flux radiant density is ¼ at 1 meters depth what is a?

$I(1)=1/4*I_0$

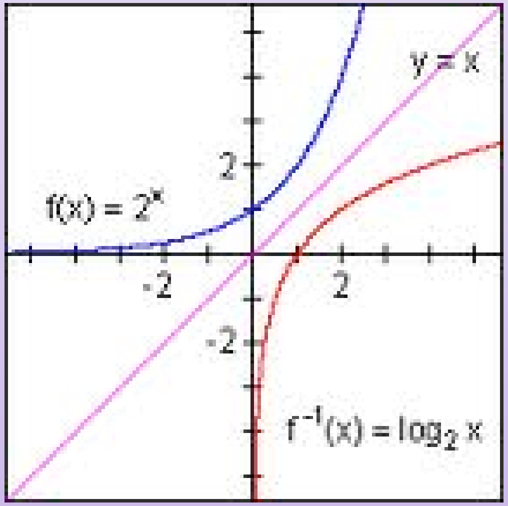
$1/4*I_0=I_0*a^1$

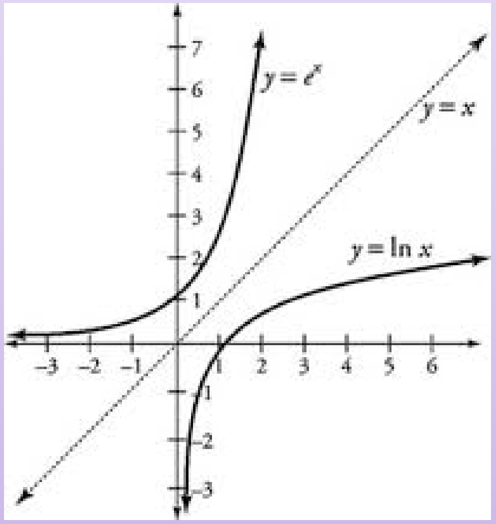
$a=1/4$

Logarithmic Functions

- Definition: A **logarithmic function** is a function of the form $f(x) = \log_a x$ where a > 0 is a fixed real number called the base of the logarithm
  - The logarithmic function is the inverse of the exponential function:

$$\log_a\left(a^x\right) = x \quad \text{and} \quad a^{\log_a x} = x$$





Logarithmic Functions

Logaritmisk form: $\log_b(y) = x$ betyder, at $b^x = y$. Logaritmen giver dig eksponenten x, som grundtallet b skal opløftes til for at få y.

For eksempel, hvis $2^3 = 8$, så er $\log_2(8) = 3$.

- **Eksponentielle funktioner** bruges til at beskrive noget, der vokser (eller falder) på en multiplikativ måde.
- **Logaritmer** er en måde at omforme noget, der vokser (eller falder) på en multiplikativ måde, så man kan måle væksten på en ny måde, hvor det vokser (eller falder) lineært.
- Dette opstår fra det faktum, at logaritmen af et produkt er summen af logaritmerne af produktets komponenter:

$$\log(ab) = \log(a) + \log(b)$$

$$\ln(ab) = \ln(a) + \ln(b)$$

Laws of Logarithms

1. $$\log_a(xy) = \log_a x + \log_a y$$

2. $$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

3. $$\log_a x^k = k\log_a x$$

4. $$\log_a a = 1 \text{ and } \log_a 1 = 0$$

5. $$\log_a x = \frac{\log x}{\log a} = \frac{\ln x}{\ln a} = \frac{\log_b x}{\log_b a}$$

Example 4.1

Solve: $$5^x = 0.23$$ use $$\log_a x^k = k\log_a x$$

Solution: $$\ln 5^x = \ln 0.23$$

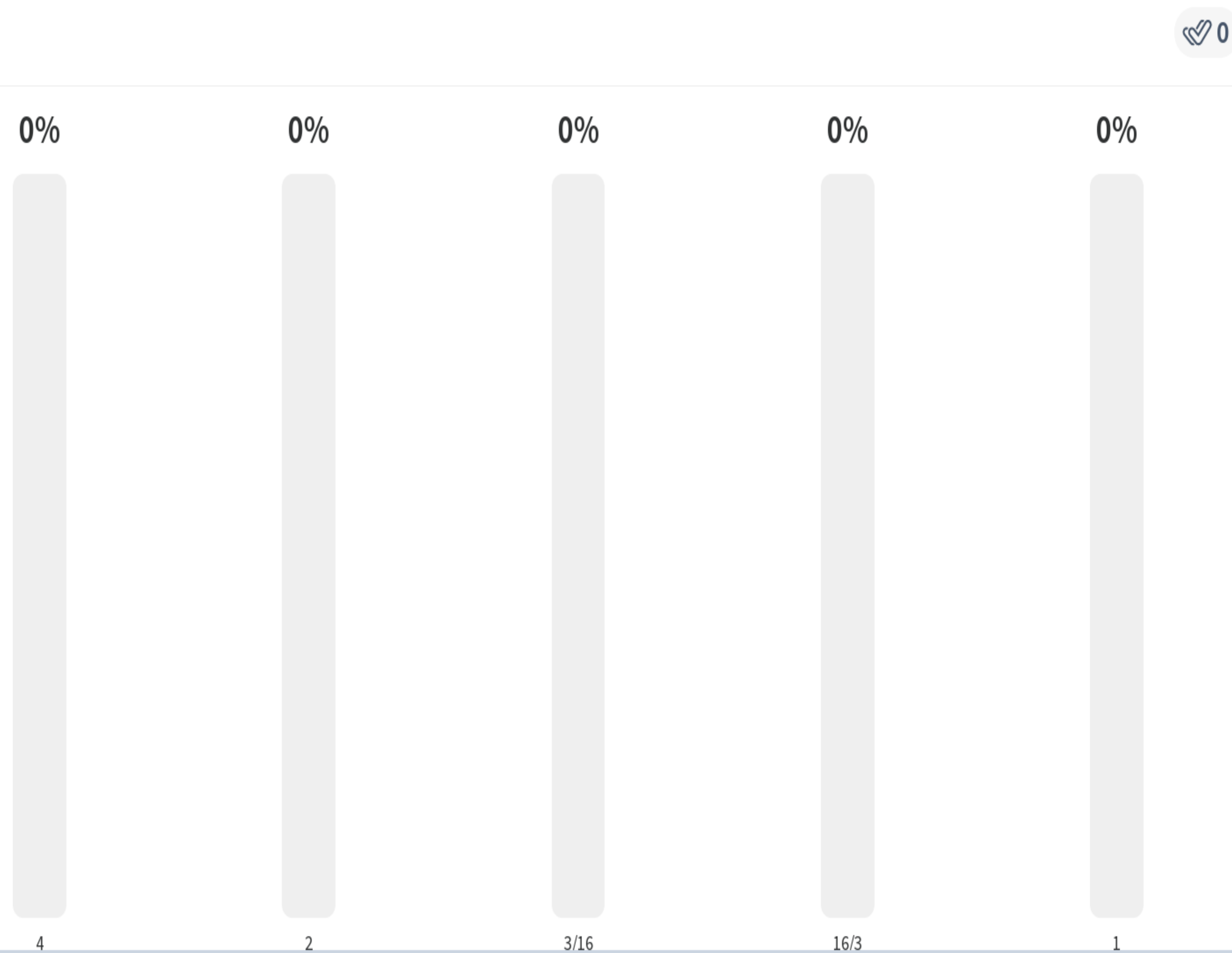
$$x\ln 5 = \ln 0.23$$

$$x = \frac{\ln 0.23}{\ln 5}$$

$$x \approx -0.91$$

Example 4.3

Solve: $$\log_4(3x) = 2$$


0
0%
0%
0%
0%
0%
4
2
3/16
16/3
1

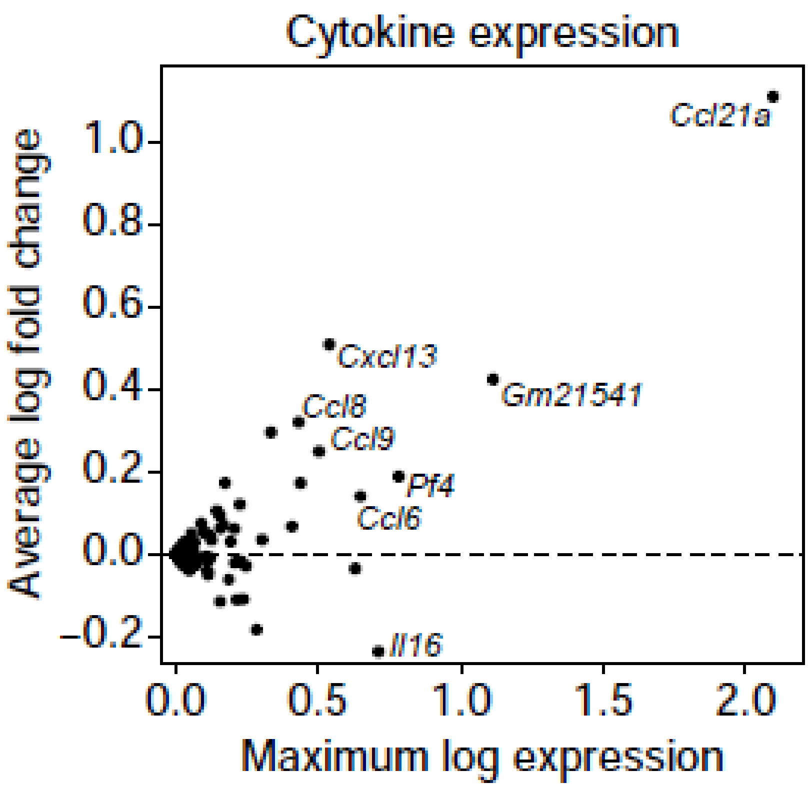
Example 4.3

Solve: $$\log_4(3x) = 2$$

Solution: $$4^{\log_4(3x)} = 4^2$$

$$3x = 4^2$$

$$x = \frac{16}{3}$$



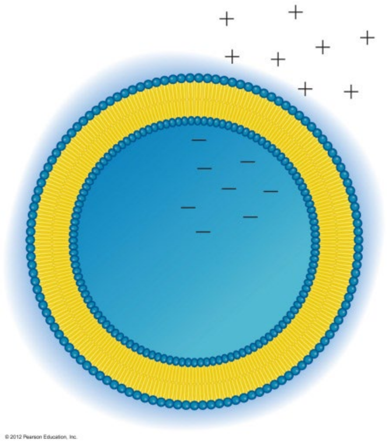
Doubling = $\log_2$ fold change of 1,
Quadrupling = $\log_2$ fold change of 2

Halving = $\log_2$ fold change of −1,
Quartering = $\log_2$ fold change of −2

Thermodynamics of transport

$$\Delta G_{\text{reaction}} = \Delta G^{\circ\prime}_{\text{reaction}} + RT \ln \frac{[C][D]}{[A][B]}$$

What does it look like for transport of a molecule " A"



$$\Delta G_{\text{transport}} = RT \ln \frac{[A_{\text{in}}]}{[A_{\text{out}}]} = 2.303\ RT \log \frac{[A_{\text{in}}]}{[A_{\text{out}}]}$$

Half-Life & Doubling Time

- Logarithms and exponentials are frequently used to calculate the half-life of radioactive substances or the doubling time of populations.
- The **half-life** of a radioactive substance is how long it takes for N amount of that substance to decay to 1/2 N amount
- The **doubling time** of a population is how long it takes N individuals in that population to reproduce and become 2N individuals

Example 4.4

- It has been shown that the amount of drug in a person's body decays exponentially
- Let C(t) denote the concentration of the drug in the bloodstream at time t (in days), and let C_0 be the initial amount of drug in the bloodstream
- Then we can write $C(t) = C_0e^{-kt}$ where $k > 0$ is known as a decay constant. The larger the value of k, the more quickly the drug decays in the bloodstream
- (a) If the drug has a half-life of 10 days, what is the value of k?

Example 4.4

- (a) A half-life of 10 days means that when t = 10 we should have 1/2 of the initial amount of drug, C_0:

$$C(10) = \frac{1}{2} C_0$$

- From the given relation we have:

$$C(10) = C_0 e^{-10k}$$

- Hence:

$$0.5C_0 = C_0 e^{-10k}$$

$$0.5 = e^{-10k}$$

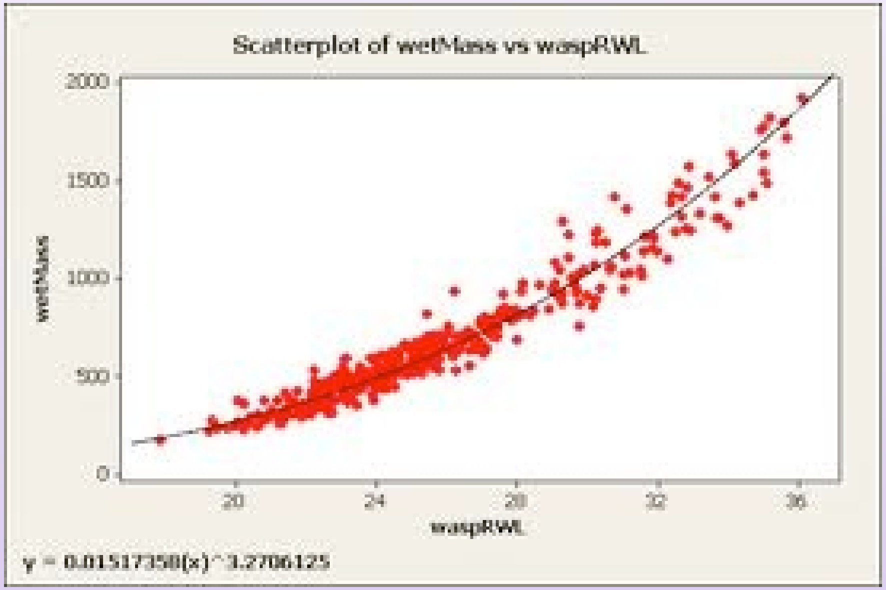
$$\ln 0.5 = \ln e^{-10k}$$

$$\ln 0.5 = -10k$$

$$0.069 \approx \frac{-1}{10} \ln 0.5 = k$$

Overview

Mass Vs. Right Wing Length
Non-Linear Model for Wasps



- Data often appear to have a strong relationship, but that relationship is not linear
- We would like to apply linear analysis, or else we would need to develop a new method to get the best fit curve
- In some situations, however, a *rescaling* of the data could transform it in such a way that the new relationship is linear

Rewriting Equations

- To see why this is the case, we begin- not with data- but with the types of curves we use to model the data
- We start with an exponential equation:

$$y = c \cdot a^x$$

$$\ln(y) = \ln\left(c \cdot a^x\right)$$

$$\ln(y) = \ln c + \ln a^x$$

$$\ln(y) = \ln c + x \ln a$$

$$\ln(y) = (\ln a)x + \ln c$$

- Let's compare the structure of this equation with the equation of a line

Rewriting Equations

$$\ln(y) = (\ln a)x + \ln c$$

$$\text{dependent variable} = \text{constant} \times \text{independent variable} + \text{constant}$$

$$y = mx + b$$

The new equation is a linear equation.

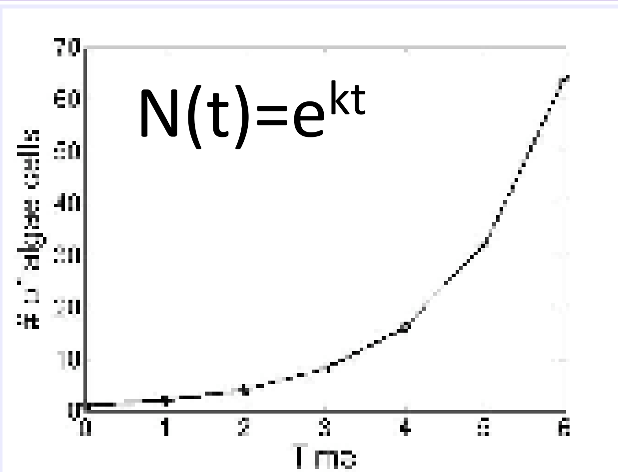
Example 4.9 (Algae Growth)

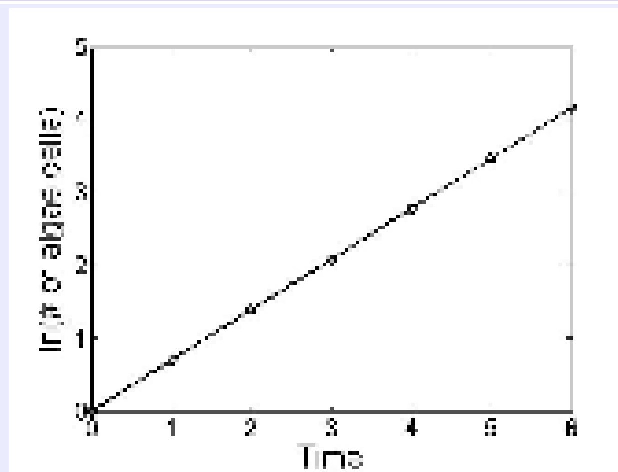
- Imagine algae growing in a Petri dish, starting from a single cell. After some time the cell will split (or else die). Then, each of the two cells will split, and so on. Thus, the number of cells in the Petri dish does not increase linearly (i.e. in an additive manner), but multiplicatively (by doubling each time).
- Suppose the doubling time is 1 day. Thus if we start off with one cell, after 1 day we will have 2 cells, after 2 days 4 cells, after 3 days 8 cells, etc.
- Let us see what happens when we rescale the y-axis data:

Example 4.9 (Algae Growth)

x-axis	y-axis	y-axis rescaled
Time t	# of cells N	$\ln N$
0	1	0
1	2	0.693
2	4	1.386
3	8	2.079
4	16	2.773
5	32	3.466
6	64	4.159

$$\ln(N(t))=\ln(e^{kt}) = kt$$





Units of measurement

Metric System

Based on the decimal system, the metric system is the common system used for scientific measurements.

TABLE 2.2 Some Prefixes for Multiples of Metric and SI Units

PREFIX	SYMBOL	BASE UNIT MULTIPLIED BY*	EXAMPLE
mega	M	$1{,}000{,}000 = 10^6$	1 megameter (Mm) $= 10^6$ m
kilo	k	$1000 = 10^3$	1 kilogram (kg) $= 10^3$ g
hecto	h	$100 = 10^2$	1 hectogram (hg) = 100 g
deka	da	$10 = 10^1$	1 dekaliter (daL) = 10 L
deci	d	$0.1 = 10^{-1}$	1 deciliter (dL) = 0.1 L
centi	c	$0.01 = 10^{-2}$	1 centimeter (cm) = 0.01 cm
milli	m	$0.001 = 10^{-3}$	1 milligram (mg) = 0.001 g
micro	μ	$0.000\,001 = 10^{-6}$	1 micrometer (μm) $= 10^{-6}$ m
nano	n	$0.000\,000\,001 = 10^{-9}$	1 nanogram (ng) $= 10^{-9}$ g
pico	p	$0.000\,000\,000\,001 = 10^{-12}$	1 picogram (pg) $= 10^{-12}$ g
femto	f	$0.000\,000\,000\,000\,001 = 10^{-15}$	1 femtogram $= 10^{-15}$ g

**The scientific notation method of writing large and small numbers (for example, 10^6 for 1,000,000) is explained in Section 2.5.*

International System of Units (SI Units)

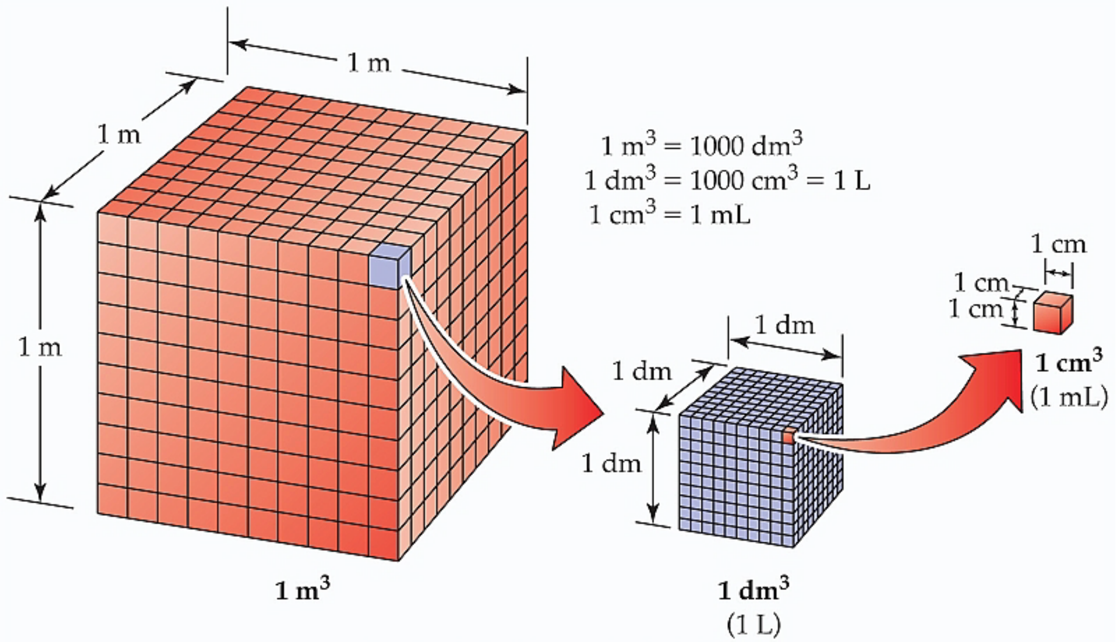
Internationally agreed upon choice of metric units; consists of base units from which all other units can be derived.

TABLE 2.1 Some SI and Metric Units and Their Equivalents

QUANTITY	SI UNIT (SYMBOL)	METRIC UNIT (SYMBOL)	EQUIVALENTS
Mass	Kilogram (kg)	Gram (g)	1 kg = 1000 g = 2.205 lb
Length	Meter (m)	Meter (m)	1 m = 3.280 ft
Volume	Cubic meter (m^3)	Liter (L)	1 m^3 = 1000 L = 264.2 gal
Temperature	Kelvin (K)	Celsius degree (°C)	See Section 2.9
Time	Second (s)	Second (s)	—

Volume: Amount of space occupied by a body.

- **SI Unit:** cubic meter (m^3)
- **Common Units:** Liter (L) or milliliter (mL) or cubic centimeter (cm^3)



Units of Concentration

- Percent volume

$$\% \text{ volume} = \frac{\text{volume solute (ml)}}{\text{volume solution (ml)}} \times 100$$

- Percent mass

$$\% \text{ mass} = \frac{\text{mass solute (g)}}{\text{mass solution (g)}} \times 100$$

$$\boxed{\text{Solution} = \text{solvent} + \text{solute}}$$

- Molarity (M) is the most common unit of concentration

  Molarity is an expression of moles/Liter of the solute.

Units of Concentration

Example 1:

- What is the percent by volume concentration of a solution in which 50.0 ml of ethanol is diluted to a volume of 250.0 ml?

- 50.0 ml x 100 = 20.0%
- 250.0 ml

Units of Concentration

Example 2:

What volume of acetic acid is present in a bottle containing 350.0 ml of a solution which measures 5.00% concentration?

0

10ml	0%
15.5ml	0%
17.5ml	0%
20ml	0%
30ml	0%

Units of Concentration

Example 2:

- What volume of acetic acid is present in a bottle containing 350.0 ml of a solution which measures 5.00% concentration?

- $\underline{\quad x \quad}$ = 0.05
- 350.0 ml
-
- x = 17.5 ml

Units of Concentration

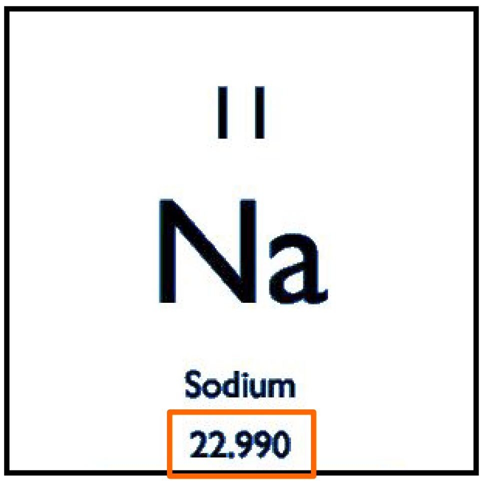
Example 3:

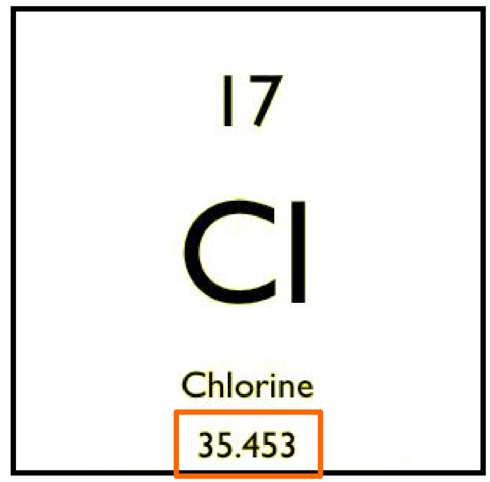
- Find the percent by mass in which 41.0 g of NaCl is dissolved in 331 grams of water.

- $\frac{41\text{ g}}{372\text{ g}} \times 100 = 11.0\%$

Units of Concentration

- The mole is the amount of substance of a system which contains as many elementary entities as there are atoms in 0.012 kilogram of carbon 12; its symbol is “mol”. About 6×10^{23} mol^{-1}
- The molar mass of a compound can be calculated by adding the molar mass of the individual elements.





22.99 + 35.45 = 58.44 g/mol

So how many grams do Y mol weigh=

Y [mol] * Molar mass [g/mol]

Making Solutions

You just calculated the molar mass of sodium chloride to be 58.44 g/mol.

To determine how to make a stock solution of sodium chloride, use the formula:

M [mol/L]x L [L] = mole fraction [mol]

g = M [mol/L]x L [L] x molar mass [g/mol]

Making Solutions

- How many grams of NaCl would you need to prepare 200.0 mL of a 5 M solution?

- g = M x L x molar mass
- g = (5mol/L) (0.2L)*(58.44g/mol)
- g = 58.44 g

Diluting Solutions

Often once you have made a stock solution, you need to dilute it to a working concentration.

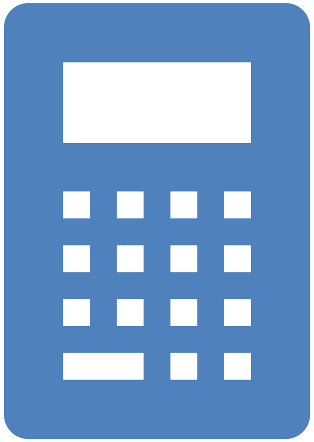
To determine how to dilute the stock solution, use the formula:

$$C_1V_1 = C_2V_2$$

C_1 – concentration of stock
C_2 - concentration of diluted solution
V_1 – volume needed of stock
V_2 – final volume of dilution

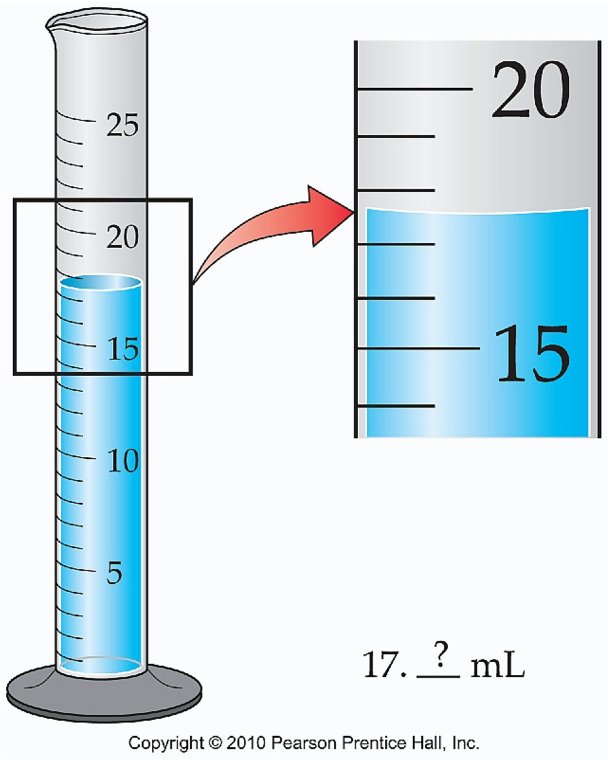
Uncertainty in Measurements

- **Exact numbers:** numbers that have a definite value.
- **Examples of exact numbers:**
  - If you buy a dozen eggs you have bought exactly 12 eggs
  - 1 kg is equal to exactly 1000 g
  - Any counted number such as number of people in a room or number of skittles in a bag
- **Inexact numbers:** numbers that do not have a definite value and contain some uncertainty. There is always uncertainty in measured quantities!



Significant Figures

All digits in a measured quantity are considered significant. The last digit of a measured quantity contains uncertainty.



Rules for Sig Figs

1) All nonzero digits are significant.
   - 457 cm has 3 sig figs
   - 2.5 g has 2 sig figs

2) Zeros between nonzero digits are significant.
   - 1007 kg has 4 sig figs
   - 1.033 g has 4 sig figs

3) Zeros to the left of the first nonzero digit are not significant. They are not actually measured but are place holders.
   - 0.0022 g has 2 sig figs
   - 0.0000022 kg has 2 sig fig

4) Zeros at the end of a number and to the right of a decimal are significant. They are assumed to be measured numbers.
   - 0.002200 g has 4 sig figs
   - 0.20 has 2 sig figs
   - 7.000 has 4 sig figs

5) When a number ends in zero but contains no decimal place, the zeros may or may not be significant. We use scientific (aka exponential) notation to specify.
   - 7000 kg may have 1, 2, 3 or 4 sig figs!

Scientific Notation

- Move the decimal behind the first nonzero digit (this will make the number between 1 and 10).

- Multiply the number by 10 to the appropriate power.

- **Examples:**
  1) 0.0001 cm = 1×10^{-4} cm
  2) 10,000 m (expressed to 1 sig fig) = 1×10^{4} m
  3) 13,333 g = 1.3333×10^{4} g
  4) 10,000 m (expressed to 3 sig figs) = 1.00×10^{4} m

NOTE: All zeros after the decimal are significant. The exponent is not counted as a sig fig.

Sig Figs In Calculations

Mult/Div: Answer must contain the same number of **sig figs** as there are in the measurement with the least number of sig figs.

$$\begin{array}{r} 2.872\underline{3} \\ \times \quad 1.\underline{6} \\ \hline 4.\underline{59568} \end{array}$$

$\Downarrow$ rounding

4.6

Add/Sub: Round answer to the same number of **decimal places** as there are in the measurement with the fewest decimal places.

$$\begin{array}{l} \,4.783\underline{2} \\ \,1.23\underline{4} \\ +\,2.0\underline{2} \\ \hline \,8.0\underline{372} \end{array}$$

$\Downarrow$ rounding

8.04

Rounding Calculations

- **Rounding:** If the left-most digit to be removed is less than 5, do not round up. If the left-most digit to be removed is greater than or equal to 5, round up.

Examples:

(6.221 cm)(5.2 cm) = 32.3492 cm^2 = 32 cm^2

(6.221 cm)(5.200 cm) = 32.3492 cm^2 = 32.35 cm^2

NOTE: Do not round until the last calculation has been performed. Rounding at each step introduces more error.

NOTE: Exact numbers (not measured numbers) are indefinitely precise and have indefinite sig figs, thus they do not ever determine the number of sig figs in a final answer! All metric conversions are exact.

NOTE: If a problem requires both addition/subtraction and multiplication/division then each rule is applied separately.

Topics

Learning goals:

1) Understand the importance of data science in life science.
    a. Why is data science important for your studies?
2) Review exponential and logarithmic functions.
    a. Why are exponential and logarithmic functions important for life science?
    b. Basic properties of exponential and logarithmic functions.
    c. Half-life and doubling time.
    d. Rescaling of data.
3) Units and concentrations.
    a. SI units.
    b. Calculations of concentrations.
4) Data representation.
    a. Significant figures.
    b. Scientific notation.